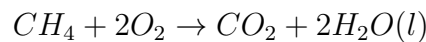


Thermal Engineering Tutorial

1 Questions

1. The products of combustion of an unknown hydrocarbon C_xH_y have the following composition as measured by an Orsat apparatus. $CO_2-8\%$, $CO-0.9\%$, $O_2-8.8\%$, $N_2-82.3\%$ Determine (a) the composition of the fuel, (b) air -fuel ratio and (c) % excess air [$C_{8.9}H_{18.5}$, AF 24, % excess air 67.2 %]
2. Determine the heat transfer per kg mol of fuel for the following reaction



The reactants and products are each at a total pressure of 100 kPa and 25 °C. [-890325 kJ]

3. Determine the adiabatic flame temperature when liquid octane at 25 °C is burned with 300% theoretical air at 25 °C in a steady flow process.
4. A gasoline engine delivers 150 kW. The fuel used is C_8H_{18} (l) and it enters the engine at 25 °C. 150 % theoretical air is used and it enters at 45 °C. The products of combustion leave the engine at 750 K, and heat transfer from the engine is 250 kW. Determine the fuel consumption per hour, if complete combustion is achieved. [39.4 kg/h]
5. Propane (g) at 25 °C and 100 kPa is burned with 400 % theoretical air at 25 °C and 100 kPa. Assume that the reaction occurs reversibly at 25 °C, that the oxygen and nitrogen are separated before the reaction takes place (each at 25 °C, 100 kPa), that the constituents in the products are separated, and that each is at 25 °C, 100 kPa. Determine the reversible work for this process. (b) If the above reaction occurs adiabatically, and each constituent in the products is at 100 kPa pressure and at adiabatic flame temperature, compute (a) the increase in entropy during combustion. (b) the irreversibility of the process and (c) the availability of the products of combustion. [47035.6 kJ/kg, ΔS - 3699.67 kJ/kg mol, I- 25001 kJ/kg, A- 22034.6 kJ/kg]
6. Liquid octane (C_8H_{18}) at 25 °C and 100 kPa and a mass flow rate of 0.57 kg/h enters a small IC engine operating at steady state. The fuel burns with air entering the engine

in a separate stream at 25 °C and 100 kPa. Combustion products exit at 670 K, 1 atm with a dry molar analysis of 11.4 % CO_2 , 2.9 % CO , 1.6 % O_2 and 84.1 % N_2 . If the engine develops power at the rate of 1kW Determine the rate of heat transfer from the engine in kW. [-4.33 kW]